Ex No: 2d

IMPLEMENTION OF FORWARD CHAINING

Aim:

To implement forward Chaining.

Procedure:

Program:

```
# Facts: The knowledge base containing information about facts
facts = {
    'a': True,
    'b': True,
    'c': False
}

# Rules: List of rules where each rule is a tuple (conclusion, conditions)
rules = [
    ('d', ['a', 'b']),  # d can be concluded if a and b are true
    ('e', ['b', 'c']),  # e can be concluded if b and c are true
    ('f', ['d', 'e'])  # f can be concluded if d and e are true
]

# Forward Chaining Function
def forward_chaining(facts, rules, goal):
```

```
    inferred = set(facts.keys())  # Initialize inferred facts with the given facts
    new_inferred = set(facts.keys())  # Start with the facts
        while new_inferred:
        current_inferred = set()  # Set to hold newly inferred facts
        for rule in rules:
            head, body = rule
            if head not in inferred and all(fact in inferred for fact in body):
                current_inferred.add(head)
                if current_inferred:
            inferred.update(current_inferred)
            new_inferred = current_inferred  # Set new inferred facts for the next iteration
        else:
            new_inferred = set()  # No new facts can be inferred

    return goal in inferred  # Check if the goal is in the inferred facts

# Example Usage
goal = 'f'

if forward_chaining(facts, rules, goal):
    print(f"The goal '{goal}' can be achieved.")
else:
    print(f"The goal '{goal}' cannot be achieved.")
```

Output:

The goal 'f' can be achieved.

The goal 'e' cannot be achieved.

The goal 'd' can be achieved.

Terminal - Ex 2d: Implementation of Forward Chaining

```
$ py code/ex2d.py

The goal 'f' can be achieved.
The goal 'e' can be achieved.
The goal 'd' can be achieved.
```